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Inoculating against fatalistic beliefs: a comparative test of narrative and non-narrative strategies

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Abstract

This study compares the effectiveness of narrative and non-narrative inoculation messages in reducing fatalistic beliefs and fostering resistance to conflicting cancer information. A two-phase randomized experiment ($N = 572$ in Phase 1; $n = 332$ in Phase 2) tested their effects on counterarguing against attack, cancer fatalism, perceived efficacy, and intentions to adopt cancer risk-reduction behaviors. Compared with control, both inoculation messages reduced cancer fatalism, enhanced perceived efficacy, and increased certain intentions to adopt certain risk-reduction behaviors. Narrative inoculation elicited more counterarguing against attack than the non-narrative counterpart. Inoculation effects also varied by baseline fatalism, functioning therapeutically by immediately reducing initially high fatalism and, more modestly, prophylactically by maintaining initially low fatalism. These findings extend inoculation theory by elucidating the role of message format in sustaining resistance to negative media effects, and by demonstrating the theory's potential dual prophylactic–therapeutic functionality in a fatalistic belief context.

The contemporary media environment exposes audiences to a vast and often conflicting body of health information. Claims about disease prevention, detection, and treatment frequently contradict one another across news and digital platforms, sowing uncertainty and confusion (Han et al., 2007). Such discord can erode trust in expert guidance and confidence in preventive action (Iles et al., 2022). In cancer communication, for example, inconsistent messages about lifestyle and cancer risks have been linked to fatalism—the pessimistic belief that cancer occurrence is predetermined and survival lies beyond personal control, a belief that in turn discourages preventive action (Iles et al., 2022; Niederdeppe et al., 2010). Fatalism thus offers a concrete context for studying resistance to conflicting information, yet the theoretical challenge runs deeper: how can communication strategies proactively equip individuals to withstand the demotivating effects of inconsistent media messages?

Inoculation theory offers one promising avenue. Since McGuire's (1961, 1964) seminal work, inoculation has served as a well-documented framework for building resistance to persuasion, forewarning audiences of an impending challenge and supplying weakened counterarguments paired with refutations. To date, however, inoculation research has largely relied on expository messages to defend specific issue attitudes against direct attack, leaving two questions underexplored. First, can inoculation extend beyond a specific attitude to confer resistance against broader belief orientations that media environments cultivate indirectly? Second, does message format matter—can narratives strengthen or weaken inoculation's core mechanisms of threat and counterarguing?

The second question is the more vexed, because the very qualities that make narratives persuasive sit in tension with how inoculation works. Narratives, a mainstay of communication practice, exert influence through transportation into story worlds and identification with characters (Cohen, 2001; Green & Brock, 2000); they characteristically overcome audiences' resistance to persuasion by reducing counterarguing (Moyer-Gusé, 2008). Yet these qualities complicate their role in inoculation, which depends on stimulating active counterarguing to prepare audiences for later persuasive attacks (Compton & Ivanov, 2013). Faced with this tension, some scholars suggest that narratives may bolster inoculation by deepening engagement with refutational content (Compton & Mason, 2020), whereas others caution that immersion may blunt the cognitive counterarguing on which inoculation's durable effects depend (Moyer-Gusé & Nabi, 2010). Despite these debates, direct empirical comparisons of narrative and non-narrative inoculation messages remain scarce.

The present study addresses these gaps by testing inoculation against the fatalistic beliefs that conflicting cancer information can induce, with the aim of advancing communication scholarship along three lines. First, it examines inoculation's capacity to protect against a broader health belief orientation shaped by conflicting media content, extending the scope of inoculation theory. Second, it provides a direct comparison of narrative and

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non-narrative inoculation messages, testing whether narrative mechanisms enhance or undermine inoculation processes. Third, it investigates whether inoculation functions prophylactically and therapeutically across individuals who differ in baseline belief, clarifying its boundary conditions and practical value. Together, these lines of inquiry aim to deepen understanding of how communication can build resistance in an environment characterized by information abundance and contradiction.

Building resistance against conflicting cancer information and fatalistic beliefs

The proliferation of health information has made encounters with contradictory messages increasingly common. Cancer communication is a telling case, in which screening guidelines, lifestyle risk factors, and treatment protocols are frequently shifting and revised. In a large national sample, more than half of respondents perceived cancer-prevention advice as inconsistent (Gollust et al., 2023). Lay audiences often struggle to reconcile such inconsistencies, which can erode trust in health authorities, reduce adherence to preventive guidelines, and breed doubt about whether prevention is worthwhile (e.g., Nagler et al., 2019).

Exposure to such conflict has been further linked to fatalism, a view that cancer is predetermined and beyond personal control. Long recognized as a barrier to prevention (Chen & Kim, 2024), fatalism can itself be cultivated by media exposure (Chen & Kim, 2024; Niederdeppe et al., 2010), particularly when audiences encounter contradictory cancer stories (e.g., Iles et al., 2022). Experimental evidence bears this out: conflicting information heightens fatalism, most sharply when the contradictions originate in evidence or from a single source, prompting people to doubt whether experts know how to prevent cancer at all (Iles et al., 2022). Such findings highlight that demotivating beliefs can arise from the structure of the information environment, not only from cultural predisposition.

Existing evidence points to a plausible mechanism underlying this effect. Conflicting information heightens perceived ambiguity, a state most people find aversive (Han et al., 2007). Ambiguity raises cognitive load as audiences strain to reconcile incompatible claims, producing frustration, decision paralysis, or disengagement (Barnwell et al., 2022). When prevention advice appears unstable, this disengagement can harden into fatalism, a defensive avoidance of the discomfort that unresolved uncertainty imposes. Through this route, conflicting information can undermine both efficacy beliefs and preventive motivation.

This study employs cancer fatalism as a focal context for addressing broader theoretical questions about resistance to adverse media effects. If conflicting information can cultivate fatalism, then effective communication strategies must build resilience proactively, fortifying audiences before such beliefs take hold. Inoculation theory provides a promising framework for this task, offering a way to forewarn individuals of impending challenges and equip them with refutations that preserve their sense of efficacy.

Inoculation theory

Inoculation theory provides a strategic framework for building resistance to persuasion by preemptively exposing individuals to weakened forms of forthcoming threats, together with refutations (McGuire, 1961, 1964). Rooted in a medical analogy, inoculation functions like a vaccine: by confronting people with a manageable threat to their beliefs, it stimulates defenses that render them resilient to stronger challenges in the future. Over six decades of research have established it as one of the most robust and versatile theories of resistance in communication, with applications spanning politics, advertising, and health (Compton, 2025), including efforts to counter misinformation (van der Linden et al., 2017).

Classic inoculation messages include two elements: forewarning and refutational preemption. Forewarning alerts recipients to a threat that their position may be challenged, whereas refutational preemption supplies weakened counterarguments and rebuttals in advance (McGuire, 1961, 1964). Recent theorizing clarifies that threat is inoculation's core mechanism: the recognition that one's position is vulnerable to challenge motivates resistance (Compton, 2025). Forewarning is one means of generating this threat that motivates resistance, while refutational preemption provides recipients with material that can stimulate counterargument to resist later attacks. Crucially, inoculation's protection is not confined to the specific counterarguments contained in the message; because it extends an umbrella of protection to novel attacks as well (Compton, 2025), its effects rest on a broader and more active resistance process than the mere recall of prepackaged rebuttals. Counterarguing is therefore an important response that threat activates within this broader resistance dynamic.

This clarification bears directly on the present study, where the anticipated challenge is not a single counterargument but the conflicting cancer information and fatalistic interpretations that undermine confidence in prevention. Building on threat as the core mechanism keeps the study grounded in the classic framework while allowing it to engage the broader informational and belief-related challenges of contemporary communication. Later work has refined how threat is operationalized, distinguishing apprehensive responses from the more motivational forms that better capture its role as a catalyst for resistance (e.g., Banas & Richards, 2017). Taken together, these developments suggest that inoculation equips individuals with both the motivation and the cognitive resources to withstand challenges to their views, in contexts where challenges may arise not only as direct counterarguments but also as uncertainty, inconsistency, and shifting interpretations.

Recent scholarship underscores inoculation's relevance to these conditions. Gretton et al. (2021) found that forewarning messages can mitigate the negative effects of changing COVID-19 guidance on trust and compliance intentions, and Nagler et al. (2023) showed that explaining the scientific process in advance helps audiences interpret conflicting information more constructively. Inoculation messages that preemptively expose politicizing tactics likewise reduce confusion and negative emotions in contested domains such as vaccination, pandemics, and climate change (e.g., van der Linden et al., 2017). Collectively,

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this work highlights inoculation's adaptability across contexts marked by inconsistency.

Important theoretical questions nonetheless remain. Most inoculation studies have targeted discrete attitudes and specific opinions about policies, products, or behaviors, leaving open whether inoculation's protective effects extend to the broader belief orientations that cumulative media exposure cultivates. Emerging research points to the potential utility of inoculation beyond discrete attitudes. For example, inoculation was found to reduce generic conspiracist beliefs (Mason et al., 2024), protect against undesired psychological states such as public speaking anxiety (Jackson et al., 2017), and address moral orientations such as resisting moral disengagement and promoting ethical behavior (Jones et al., 2026).

Even so, inoculation's application to health-related interpretive schemas remains limited. Cancer fatalism is one such schema: it reflects a durable orientation that shapes how individuals interpret uncertainty about cancer prevention. Extending inoculation here tests whether threat and counterarguing can fortify resistance not to a discrete persuasive attack but to an orientation built up through repeated exposure to conflicting health information. If effective, the approach would demonstrate inoculation's capacity to protect belief structures that govern long-term engagement with health messages.

Moreover, inoculation is especially well suited to countering fatalistic responses to conflicting information. Fatalism is linked to low efficacy and functions as a form of defensive avoidance when individuals feel incapable of effective action (Witte et al., 2001). Inoculation addresses this directly: its explicit refutations of fatalistic counterarguments implicitly bolster efficacy, affirming that prevention remains possible and worthwhile. By pairing a manageable threat (the warning that conflicting information may pull one toward fatalism) with a strong efficacy appeal grounded in evidence-based refutation, inoculation encourages adaptive information processing in place of avoidance, positioning it to forestall the demotivating effects of conflicting cancer information.

Despite this fit, inoculation has rarely been applied to fatalistic beliefs. By testing whether its mechanisms hold in this domain, the present study extends the theory to one of the defining challenges of communication in an age of information abundance and contradiction. The following hypothesis is thus proposed:

H1: Compared to the control group, inoculated participants will show: (a) greater counterarguing against attack, (b) reduced cancer fatalism, (c) higher perceived efficacy in cancer prevention, and (d) greater intentions to adopt cancer risk-reduction behaviors.

Narrative inoculation messages

Narratives have long been recognized as powerful tools of persuasion. As connected sequences of events featuring characters and outcomes, narratives influence audiences not through explicit claims but by drawing them into story worlds and characters' perspectives. Research on narrative persuasion shows that such immersion can shape beliefs by reducing resistance and modeling behaviors in ways expository messages do not (Green & Brock, 2000; Moyer-Gusé, 2008). These qualities have made narratives prominent in communication campaigns, advocacy, and entertainment-education. Two mechanisms are central to narrative influence. Transportation is the state of absorption in a story world, in which attention and emotion shift toward its unfolding events (Green & Brock, 2000); identification is the adoption of a character's perspective, such that the audience experiences the story through that character's thoughts, feelings, and choices (Cohen, 2001). Together, these processes make a message feel less overtly persuasive and heighten acceptance of the ideas it conveys.

Narratives and inoculation: revisiting a theoretical tension

Although narratives are widely valued for their persuasive potential, their role in inoculation remains unsettled. Scholarship suggests a dual role (Compton & Mason, 2020). Narrative features such as characters' personal experiences can make threat and refutation more concrete and memorable, potentially strengthening resistance (Lee et al., 2016); yet story immersion may simultaneously suppress the counterarguing on which inoculation depends, as absorbed audiences grow less inclined to scrutinize claims or generate opposing thoughts (Green & Brock, 2000; Moyer-Gusé, 2008; Moyer-Gusé & Nabi, 2010). This dual role creates an apparent tension with inoculation's core function: inoculation protects attitudes by supplying weakened counterpositions and refutations that recipients can mobilize when their attitudes are later challenged (McGuire, 1964), so counterarguing is integral to its operation—yet the very immersion that makes narrative compelling may blunt that process.

This tension, however, may reflect an overlooked conceptual distinction. The counterarguing that narratives suppress is directed at the message immediately encountered at that moment, whereas inoculation concerns resistance to a later attack, not to the inoculation message itself. Seen this way, narrative immersion need not conflict with inoculation. If lowered resistance to the inoculation message helps recipients absorb and retain its refutational content, their defenses may be reinforced when the attack arrives. Existing evidence speaks only partly to this possibility. Inoculation messages occasionally incorporate narrative elements, such as dramatized refutations, to enhance vividness and engagement (Compton & Mason, 2020), but message format has seldom been treated as a design feature of inoculation in its own right. More often, narrative and inoculation are examined as separate strategies: Niederdeppe et al. (2015), for example, found that narrative messages increased policy support immediately after exposure, whereas inoculation messages better protected attitudes a week later, and Geegan et al. (2023) found greater transportation and identification for narrative messages but stronger delayed resistance for inoculation messages. Informative as they are, such comparisons cannot show whether inoculation itself operates differently when its core components are delivered in narrative versus expository form.

Toward narrative inoculation

Recent work suggests that the facilitative side of this dual-role account may, under some conditions, emerge in practice. Biddlestone et al. (2023), for instance, found that narrative-based

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inoculation increased resistance to conjunction errors, reducing conspiracy beliefs and improving truth discernment. Narratives, on this evidence, may instantiate rather than obstruct inoculation, furnishing a concrete vehicle through which threat and refutation are experienced and sustained. Several features of narrative make this theoretically plausible. By embedding counterarguments in character-driven episodes, narratives can dramatize faulty reasoning, model refutations, and make abstract claims more concrete. Compared with expository inoculation, narrative formats may reduce psychological distance by situating threat and refutation within relatable characters and experiences (Compton & Mason, 2020). In doing so, they may make both vulnerabilities to challenge and the logic of resistance more accessible. Moreover, some evidence suggests that narratives can evoke anticipatory threat even when forewarning is less explicit (Niederdeppe et al., 2015), indicating that narrative form may activate parts of the inoculation process.

Nevertheless, evidence from a direct comparison of message formats for delivering inoculation's core components remains sparse. The present study addresses this gap by comparing narrative and non-narrative inoculation messages that hold the inoculation elements constant while varying whether they are delivered in story-based or expository form. We further examine transportation and identification as potential mediators, since narrative engagement may not operate uniformly in resistance contexts (Compton & Mason, 2020). This design permits a test not only of whether narrative inoculation works, but also of whether these narrative mechanisms account for any differences between formats. The following research questions are therefore posed:

RQ1: How will narrative inoculation differ from non-narrative inoculation in influencing (a) counterarguing against attack, (b) cancer fatalism, (c) perceived efficacy in cancer prevention, and (d) intentions to adopt cancer risk-reduction behaviors?

RQ2: How will (a) identification and (b) transportation mediate the effects of narrative inoculation (vs. non-narrative inoculation)?

Prophylactic and therapeutic inoculation

A recent development in inoculation theory extends its function to "treat" individuals who already hold beliefs opposed to the advocated position (Compton, 2020). Traditionally, inoculation has been prophylactic, akin to vaccination, designed to build resistance in healthy individuals before contracting the pathogen. Classic studies accordingly screened participants beforehand to confirm an initial position and then delivered messages that reinforced it (e.g., Ivanov et al., 2009). The baseline attitude served as a precondition rather than a variable of interest—what mattered was that a defensible position existed prior to inoculation.

In this paradigm, inoculation defends an established attitude against persuasive attack. Its forewarning component presumes a preexisting belief to protect; absent such an anchor, perceived threat weakens, and so does the motivation to defend that belief. This dependence on an initial attitudinal position has long been recognized as a boundary condition of the theory. Prophylactic inoculation thus targets recipients in a "healthy" cognitive state, aiming to prevent attitude change. In the context of fatalism, prophylactic inoculation would serve to maintain non-fatalistic beliefs: it would work to keep non-fatalists non-fatalistic. Yet health interventions unfold in settings where audiences differ widely in their prior beliefs, raising a critical question: if recipients already hold beliefs contrary to the inoculation message, can inoculation shift their position in the advocated direction?

To address this, scholars have turned to therapeutic inoculation, the analogue of therapeutic vaccination in medicine, administered after an individual has already been "infected" with a harmful belief (Eggermont, 2009). This line of research asks whether inoculation can correct or mitigate already-formed beliefs rather than merely prevent their formation. Wood (2007) was among the first to test this possibility directly. In this work on agricultural biotechnology, inoculation strengthened positive attitudes toward the advocated position, particularly among those initially opposed. Notably, participants with neutral or opposing attitudes perceived greater threat from the inoculation message than did supportive participants, hinting at therapeutic inoculation's promise.

Subsequent research provides evidence that inoculation can operate across the attitudinal spectrum. Ivanov et al. (2017) found that it shifted attitudes even among those initially neutral or opposed to certain tourist destinations, with the shifts persisting after counter-attitudinal attacks. Similarly, van der Linden et al. (2017) showed that inoculation messages emphasizing the scientific consensus on climate change both reinforced pro-science attitudes and reduced skepticism among those prone to doubt. Mason et al. (2024) extended the pattern to conspiracy beliefs, documenting effects that were at once prophylactic and therapeutic: inoculation bolstered resistance among those with lower baseline conspiracist beliefs while reducing such beliefs among those with higher predispositions.

These findings carry practical implications for fatalism interventions, where the goal is not only to shield non-fatalists from adopting fatalistic beliefs but also to correct those who already hold them. In today's fragmented and often contradictory information environment, anyone may be vulnerable to fatalistic thinking, making it essential that real-world message strategies engage diverse audiences simultaneously, functioning prophylactically and therapeutically, to benefit all recipients regardless of baseline positions. As Compton and Ivanov (2013) note, inoculation can do both: reinforcing supportive attitudes while shifting neutral or opposed positions toward the advocated direction. This dual applicability positions inoculation as a generalizable strategy for building resilience across the full spectrum of belief levels. The following hypotheses are therefore proposed:

H2: Among individuals lower in baseline cancer fatalism, inoculated participants will report less belief change counter to the advocated position (i.e., maintained low levels of cancer fatalism) than control participants.

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H3: Among individuals higher in baseline cancer fatalism, inoculated participants will report greater belief change toward the advocated position (i.e., reduced cancer fatalism) than control participants.

Method

This study used a two-phase online experiment with Singaporean adults recruited through Dynata's online panel. Asian populations are considered more strongly influenced by cancer fatalism than other groups (Chen & Kim, 2024). Among Asian societies, Singapore offers a particularly suitable setting: its multiethnic population, comprising primarily Chinese, Malay, and Indian communities, provides a rich cultural backdrop for studying cancer fatalism. Its relatively high prevalence of fatalistic beliefs and low uptake of preventive behaviors (Kim & Lwin, 2017), together with limited economic barriers to health promotion and high rates of Internet penetration and news consumption, make it well suited for examining how communication shapes cancer control. In total, 572 participants completed Phase 1 of the experiment, and 332 completed the follow-up two weeks later (58.0% retention). Sociodemographic characteristics are reported in Table S1.

Stimuli and pre-test

All three Phase 1 stimuli began with an overview of the cancer information landscape; the two inoculation messages then added inoculation components. The non-narrative message used a conventional expository format, forewarning participants about the threat posed by conflicting information and its potential to foster cancer fatalism, then preemptively refuting anticipated counterarguments with evidence affirming the value of cancer prevention. The narrative message conveyed the same content through a first-person story. Conflicting cancer information in the media initially left the protagonist confused and pessimistic (forewarning), but, guided by a doctor, the protagonist learned to verify claims and adopt a healthier lifestyle (refutational preemption). Ultimately, the protagonist felt less fatalistic, more alert to conflicting information, more confident about reducing cancer risk, and healthier overall.

A pre-test ($N = 210$) confirmed that, as intended, the inoculation messages elicited greater perceived threat than the control message. Participants were randomly assigned to one of the three conditions, and a one-way ANOVA revealed a significant difference in elicited threat, $F(2, 207) = 4.23$, $p = .016$, $\eta^2 = .04$. Post hoc comparisons indicated that both the non-narrative inoculation ($n = 71$, $M = 5.81$, $SD = 0.69$, $SE = .13$, $p = .049$) and the narrative inoculation ($n = 69$, $M = 5.85$, $SD = 0.68$, $SE = .13$, $p = .024$) elicited significantly higher threat than the control group ($n = 70$, $M = 5.51$, $SD = 0.87$), validating the manipulation.

The attack messages used to challenge participants' beliefs featured conflicting information about cancer prevention. Such information refers to messages about a single health behavior that present different or opposing claims, or that contain logically inconsistent assertions, thereby creating uncertainty about what is accurate and what course of action to take (Nagler & LoRusso, 2017). The present attack messages were designed to embody this form of informational and decisional conflict.

In Phase 2, participants read an attack message presenting conflicting scientific evidence on one of two cancer-prevention themes (red meat consumption or hair dye use). Each message first cited research suggesting a cancer risk, then contrasted it with conflicting findings, shifting conclusions, or alternative explanations. This juxtaposition of possible risk and unresolved uncertainty created ambiguity about the behavioral implications, mirroring the conflicting cancer information people routinely encounter in public discourse.

Each attack message also closed with brief fatalistic statements. These were intended not as the central persuasive claim but as a simulation of natural "antigenic exposure," a feature central to this study and consistent with the medical analogy underlying inoculation theory (McGuire, 1961, 1964). They reflected the interpretive leap that often follows scientific uncertainty, whereby unresolved evidence invites resignation rather than action.

Although theme comparisons were not a focus of the study, using both themes added variety and strengthened the robustness of the inoculation test. Independent-samples t-tests revealed no significant differences between themes in perceived message credibility, source credibility, perceived efficacy, counterarguing, or cancer fatalism. Given this statistical equivalence, the two conditions were combined for analysis. All stimuli are available in Supplementary Appendix A.

Main experiment procedures

The two-phase experiment was conducted online. In Phase 1, participants completed baseline measures of cancer fatalism and issue involvement before being randomly assigned to read a narrative inoculation message, a non-narrative inoculation message, or a control message. Immediately after exposure, their threat perceptions, identification, transportation, perceived efficacy, behavioral intentions, and postintervention fatalism were assessed. Two weeks later, in Phase 2, participants received an attack message containing conflicting cancer information; their counterarguments were recorded, followed by final assessments of cancer fatalism, perceived efficacy, and intentions to adopt cancer risk-reduction behaviors.

Measurements

Table S2 reports the descriptive statistics and correlations for study variables, and Table S3 presents the full wording of all measurement items.

Elicited threat

Elicited threat was assessed separately for conflicting cancer information and cancer fatalism. Consistent with recent developments in inoculation research, threat was operationalized as two related but distinct perceptions: apprehensive threat and motivational threat (e.g., Banas & Richards, 2017). Apprehensive threat captures the extent to which a challenge is perceived as unsettling or threatening, and motivational threat reflects the extent to which that challenge activates a defensive motivation and readiness to resist (Banas & Richards, 2017). After Phase 1 exposure, apprehensive threat was measured with five semantic differential items on 7-point scales (e.g., 1 = *nonthreatening*,

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7 = *threatening*), with reliability of α = .94 for both conflicting cancer information and cancer fatalism. Motivational threat was measured with four agreement-scale items (e.g., "I want to defend my current beliefs in cancer prevention"), yielding α = .81 for conflicting cancer information and α = .83 for cancer fatalism. Both scales were adopted from Banas and Richards (2017).

Cancer fatalism

Cancer fatalism was measured at baseline, postintervention, and postattack using a 20-item scale adapted from Shen et al. (2009) for the cancer context. Rated on 7-point agreement scales, the items captured three dimensions of cancer fatalism: predetermination (e.g., "If someone is meant to get cancer, they will get it no matter what they do"), luck (e.g., "Whether I will get cancer is a matter of luck"), and pessimism (e.g., "There is really no way I can prevent cancer"). Reliabilities across the three assessments were α_{baseline} = .95, $\alpha_{\text{postintervention}}$ = .97, α_{final} = .96.

Counterarguing against attack

Counterarguing against the attack was measured with five items adopted from Niederdeppe et al. (2015) on a 7-point agreement scale (e.g., "I felt like I wanted to 'argue back' with what was being said in the article"; α = .92).

Perceived efficacy

Perceived efficacy was measured after Phase 1 and Phase 2 message exposure with four items from the Risk Behavior Diagnosis Scale (Witte et al., 2001), adapted for the cancer prevention context. The scale assessed self-efficacy and response efficacy on 7-point agreement scales ($\alpha_{\text{Phase 1}}$ = .85, $\alpha_{\text{Phase 2}}$ = .86).

Intentions to adopt cancer risk-reduction behaviors

After Phase 1 and Phase 2 message exposure, intentions were assessed for four modifiable behaviors most commonly identified in health guidance as relevant to cancer risk reduction: fruit and vegetable consumption, physical activity, smoking avoidance, and alcohol avoidance. Each was measured with a single item asking how likely participants were to engage in the behavior over the next two weeks (1 = *extremely unlikely*, 7 = *extremely likely*).

Identification and transportation

Identification and transportation were captured after Phase 1 message exposure with scales adopted from Watts et al. (2025), whose wording suited both narrative and non-narrative messages. Identification was assessed with six items (e.g., "While reading, I felt I could really get inside the author's head"; α = .93), and transportation with three items (e.g., "I could picture myself dealing with the situations described in the article"; α = .87), all on 7-point agreement scales.

Issue involvement

Following prior inoculation research (e.g., Ivanov et al., 2009), issue involvement was measured before Phase 1 message exposure and included as a covariate. Because it can shape resistance and counterarguing (Pfau et al., 2004), controlling for its baseline level helps isolate inoculation effects. Participants rated six semantic differential items in response to the prompt "To me, cancer prevention is …" on a 7-point scale (e.g., 1 = *unimportant*, 7 = *important*). Reliability was α = .92.

Results

Analytical approach

All analyses were conducted in SPSS v.27. Chi-square tests revealed no significant differences across conditions in age, gender, ethnicity, education, household income, or personal and family cancer history (see Table S1), consistent with successful randomization; these sociodemographic variables were therefore not included as covariates.

To test the comparative effects of the experimental conditions, separate one-way multivariate analyses of covariance (MANCOVA) were conducted for the Phase 1 and Phase 2 outcomes, using the respective analytic samples (N = 572 in Phase 1; n = 332 in Phase 2). Baseline cancer fatalism and issue involvement served as covariates, and condition (control, non-narrative inoculation, or narrative inoculation) served as the between-subjects factor. Two orthogonal planned Helmert contrasts followed up the omnibus effect of condition. The first contrast (C1) compared the control condition with the two inoculation conditions combined, directly testing H1 that inoculation would yield more favorable outcomes than control, and the second contrast (C2) compared the two inoculation conditions, addressing RQ1 on whether the narrative and non-narrative formats differed in their effects.

Mediation and moderation analyses used the PROCESS macro for SPSS. To address RQ2, the mediation model was estimated on the inoculation-only subsample (non-narrative = 1, narrative = 2), since it sought to explain differences between the two inoculation formats rather than the overall inoculation-versus-control effect. Parallel mediation was tested with Model 4, specifying identification and transportation as simultaneous mediators and issue involvement as a covariate; indirect effects were estimated via ordinary least squares (OLS) regression with 5,000 bootstrap resamples and 95% percentile confidence intervals. To test whether inoculation effects varied with baseline fatalism (H2 and H3), moderation was examined using Model 1, with baseline fatalism and issue involvement mean-centered. Experimental condition was coded control = 1 versus inoculation = 2, and significant interactions were probed using the Johnson–Neyman technique (Johnson & Fay, 1950).

Omnibus results

For the Phase 1 outcomes, there was a significant multivariate effect of condition, Wilks' Λ = .90, $F(24, 1{,}112) = 2.59$, $p < .001$, $\eta_p^2 = .05$. The multivariate effects were also significant for baseline cancer fatalism, Wilks' Λ = .41, $F(12, 556) = 66.88$, $p < .001$, $\eta_p^2 = .59$, and issue involvement, Wilks' Λ = .82, $F(12, 556) = 9.88$, $p < .001$, $\eta_p^2 = .18$.

For the Phase 2 outcomes, the multivariate test likewise showed significant effects of condition, Wilks' Λ = .91, $F(14, 642) = 2.25$, $p = .005$, $\eta_p^2 = .05$, as well as baseline cancer fatalism, Wilks' Λ = .47, $F(7, 321) = 50.75$, $p < .001$, $\eta_p^2 = .53$, and issue involvement, Wilks' Λ = .84, $F(7, 321) = 9.01$, $p < .001$, $\eta_p^2 = .16$.

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Manipulation checks

Univariate tests showed significant effects of condition on all four threat measures: apprehensive threat toward conflicting cancer information, $F(2, 567) = 3.44$, $p = .033$, $\eta^2p = .01$, and cancer fatalism, $F(2, 567) = 4.84$, $p = .008$, $\eta^2_p = .02$; motivational threat toward conflicting cancer information, $F(2, 567) = 9.51$, $p < .001$, $\eta^2p = .03$, and cancer fatalism, $F(2, 567) = 10.80$, $p < .001$, $\eta^2p = .04$. Helmert contrast C1 (see Table 1) indicated that, relative to control, participants in the inoculation conditions reported significantly higher apprehensive threat toward both conflicting cancer information (estimate = −.28, $SE = .11$, $p = .012$) and cancer fatalism (estimate = −.30, $SE = .10$, $p = .004$), as well as higher motivational threat toward both conflicting cancer information (estimate = −.35, $SE = .08$, $p < .001$) and cancer fatalism (estimate = −.38, $SE = .08$, $p < .001$). Helmert contrast C2 showed no significant differences between the two inoculation conditions on any threat measure (all $ps > .24$), indicating that narrative and non-narrative formats were equivalent in eliciting threat (also see Table S4). In sum, inoculation strategies elevated threat perceptions relative to control as intended.

Traditional inoculation outcomes

Univariate analyses revealed significant effects of condition on postintervention fatalism, $F(2, 567) = 6.12$, $p = .002$, $\eta^2_p = .02$; postattack fatalism, $F(2, 327) = 5.10$, $p = .007$, $\eta^2_p = .03$; postintervention efficacy, $F(2, 567) = 4.67$, $p = .010$, $\eta^2_p = .02$; postattack efficacy, $F(2, 327) = 7.11$, $p < .001$, $\eta^2_p = .04$; and only Phase 1 alcohol avoidance intention, $F(2, 567) = 4.13$, $p = .017$, $\eta^2_p = .01$, among the behavioral intentions. The effect on counterarguing against the attack message was not significant, $F(2, 327) = 2.62$, $p = .075$, $\eta^2_p = .02$.

Helmert contrast C1 (see Table 1) showed that, compared with control, inoculated participants reported lower fatalism and higher perceived efficacy at both postintervention (fatalism: estimate = .25, $SE = .07$, $p < .001$; efficacy: estimate = −.25, $SE = .08$, $p = .002$) and postattack (fatalism: estimate = .28, $SE = .09$, $p = .002$; efficacy: estimate = −.42, $SE = .11$, $p < .001$), supporting H1b and H1c. The contrast was not significant for counterarguing against attack (estimate = .11, $SE = .14$, $p = .438$); H1a was not supported. Inoculated participants also reported stronger intentions to consume fruit and vegetables (estimate = −.25, $SE = .11$, $p = .029$) and avoid alcohol (estimate = −.35, $SE = .12$, $p = .005$) in Phase 1, and to engage in physical activity in Phase 2 (estimate = −.39, $SE = .17$, $p = .019$), partially supporting H1d.

Narrative vs. non-narrative inoculation

Helmert contrast C2 (see Table 1) revealed no significant differences between the non-narrative inoculation and narrative inoculation conditions in identification (estimate = −.13, $SE = .10$, $p = .205$) or transportation (estimate = −.06, $SE = .10$, $p = .547$). Although the combined inoculation conditions did not differ significantly from control in counterarguing, participants in the narrative inoculation condition reported greater counterarguing against the subsequent attack than those in the non-narrative inoculation condition (estimate = −.36, $SE = .17$, $p = .031$), addressing RQ1a. No significant differences between the narrative and non-narrative inoculation conditions emerged for cancer fatalism (RQ1b), perceived efficacy (RQ1c), or behavioral intentions (RQ1d). Covariate-adjusted Dunnett comparisons showed that each format produced a similar pattern of significant favorable outcomes relative to control (see Table S4). Neither format consistently outperformed the other.

For RQ2, the mediation model showed that narrative inoculation generated higher counterarguing against the attack ($b = .37$, $SE = .16$, $t = 2.25$, $p = .026$, 95% CI [.05, .69]), consistent with Helmert contrast C2. However, none of the indirect effects through identification or transportation were significant; all bootstrap confidence intervals included zero. Thus, neither narrative engagement mechanism mediated the effect on counterarguing.

Prophylactic vs. therapeutic inoculation

Postintervention change in cancer fatalism (postintervention minus baseline) was regressed on experimental condition (control vs. inoculation), baseline fatalism, and their interaction, controlling for issue involvement (see Table S5). The interaction was significant ($b = -.15$, $SE = .06$, $t = -2.31$, $p = .021$, 95% CI [−.27, −.02]), indicating that the inoculation effect strengthened as baseline fatalism increased. The Johnson–Neyman technique identified a transition point of 3.49. The two conditions did not differ significantly below this value of baseline fatalism, whereas inoculation significantly reduced fatalism relative to the control condition above it, encompassing approximately 74% of the sample (see Table S6 and Figure 1).

To interpret this region in relation to H2 and H3, the midpoint of the 7-point scale (4 = neither agree nor disagree) served as a reference. Following prior research that treated a neutral response as the boundary between preexisting positions (e.g., Ivanov et al., 2017; Wood, 2007), scores of 4 or below denoted lower baseline fatalism, whereas scores above 4 denoted higher baseline fatalism. These ranges correspond, respectively, to the comparatively "healthy" state relevant to prophylactic inoculation and the "already affected" state relevant to therapeutic inoculation. As expected, participants in the lower-fatalism range ($n = 248$, $M = 3.05$, $SD = 0.79$) scored significantly lower than those in the higher-fatalism range ($n = 324$, $M = 4.89$, $SD = 0.63$), $t(465) = -30.09$, $p < .001$. Because the Johnson–Neyman transition point (3.49) fell below 4, the significant effect included the entire higher-fatalism range, supporting H3 at postintervention and indicating an immediate therapeutic effect. It also extended into the upper portion of the lower-fatalism range (3.49–4.00), consistent with a prophylactic function of preserving or reinforcing comparatively low fatalism. Because the effect did not extend below 3.49, H2 received partial support at postintervention.

A parallel analysis examined postattack change in cancer fatalism (postattack minus baseline). The interaction was not significant ($b = -.11$, $SE = .08$, $t = -1.35$, $p = .176$, 95% CI [−.27, .05]). Exploratory Johnson–Neyman probing nevertheless identified a similar transition point of 3.37 (see Table S7 and Figure 1), with the estimated inoculation effect being nonsignificant below this baseline fatalism value and significant above it. This region

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Table 1 Results of Helmert contrasts.

	Control		Non-narrative inoculation		Narrative inoculation		Control vs. inoculation			Non-narrative vs. narrative inoculation		
Dependent variables	*M*	*SD*	*M*	*SD*	*M*	*SD*	Estimate	*SE*	*p*	Estimate	*SE*	*p*
Manipulation checks												
Apprehensive threat (CCI)[a]	5.22	1.32	5.54	1.24	5.42	1.24	−.28	.11	**.012**	.09	.13	.472
Apprehensive threat (cancer fatalism)[a]	5.32	1.24	5.55	1.22	5.64	1.11	−.30	.10	**.004**	−.14	.12	.243
Motivational threat (CCI)[a]	4.71	1.00	5.03	0.85	5.06	0.88	−.35	.08	**<.001**	−.06	.09	.503
Motivational threat (cancer fatalism)[a]	4.67	1.04	5.05	0.88	5.04	0.92	−.38	.08	**<.001**	−.03	.09	.747
Narrative mechanisms												
Identification[a]	4.55	1.14	4.86	0.90	4.95	0.95	−.37	.09	**<.001**	−.13	.10	.205
Transportation[a]	4.63	1.18	4.94	0.93	4.96	0.96	−.34	.09	**<.001**	−.06	.10	.547
Inoculation outcomes												
Counterarguing against the attack[b]	3.85	1.28	3.54	1.22	3.94	1.18	.11	.14	.438	−.36	.17	**.031**
Cancer fatalism (post-intervention)[a]	4.25	1.22	3.95	1.30	3.99	1.21	.25	.07	**<.001**	.04	.08	.657
Cancer fatalism (final)[b]	4.35	1.13	3.96	1.20	4.12	1.05	.28	.09	**.002**	.04	.10	.738
Perceived efficacy (Phase 1)[a]	5.06	1.11	5.34	0.91	5.29	0.90	−.25	.08	**.002**	.00	.09	.996
Perceived efficacy (Phase 2)[b]	4.76	1.14	5.17	0.98	5.18	0.97	−.42	.11	**<.001**	−.10	.13	.451
Intention—F&V consumption (Phase 1)[a]	5.18	1.40	5.38	1.28	5.47	1.19	−.25	.11	**.029**	−.13	.13	.324
Intention—F&V consumption (Phase 2)[b]	5.04	1.59	5.17	1.33	5.37	1.26	−.24	.16	.137	−.29	.18	.115
Intention—physical activity (Phase 1)[a]	5.07	1.54	5.27	1.24	5.29	1.42	−.22	.12	.075	−.07	.14	.635
Intention—physical activity (Phase 2)[b]	4.77	1.61	5.17	1.44	5.16	1.46	−.39	.17	**.019**	−.10	.19	.599
Intention—smoking avoidance (Phase 1)[a]	6.03	1.53	6.22	1.31	6.24	1.14	−.22	.12	.065	−.07	.13	.624
Intention—smoking avoidance (Phase 2)[b]	6.14	1.50	6.37	1.24	6.30	1.18	−.21	.15	.151	.01	.17	.961
Intention—alcohol avoidance (Phase 1)[a]	5.51	1.61	5.91	1.21	5.78	1.40	−.35	.12	**.005**	.08	.14	.569
Intention—alcohol avoidance (Phase 2)[b]	5.59	1.77	5.73	1.54	5.66	1.47	−.11	.18	.537	.00	.21	.986

Note. Contrast estimates are adjusted for baseline cancer fatalism and issue involvement. Significant *p*-values are in bold. CCI = conflicting cancer information; F&V = fruit and vegetables.

[a]$N = 572$.

[b]$n = 332$.

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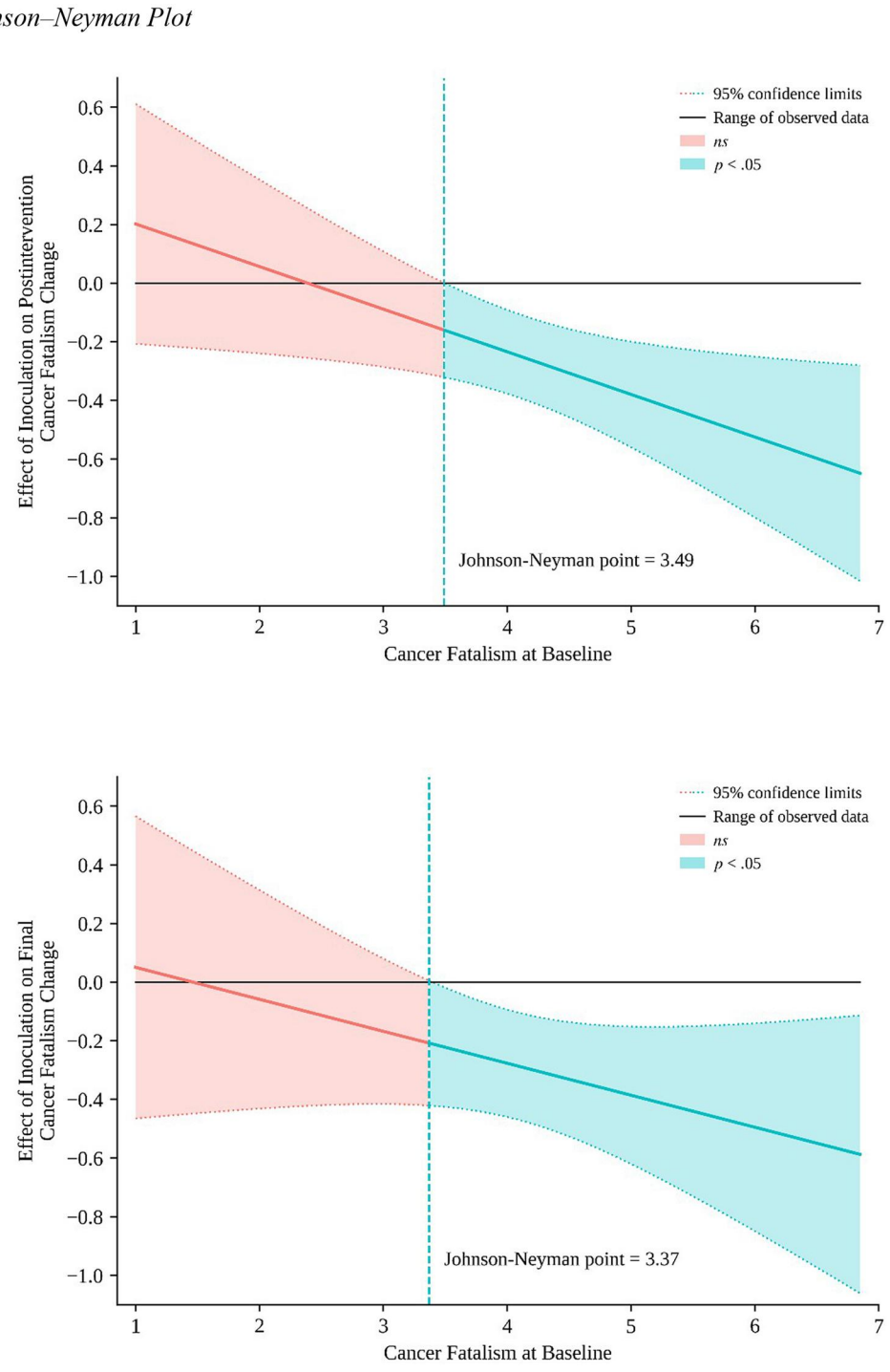


Figure 1 Johnson–Neyman plot.

again included the entire higher-fatalism range and the upper portion of the lower-fatalism range. Given the nonsignificant interaction, however, this pattern did not provide confirmatory evidence that the therapeutic and partial prophylactic effects persisted after the attack. In summary, the hypotheses received support only at postintervention: H2 received partial support, whereas H3 was supported. Descriptive statistics for fatalism changes across conditions and phases by baseline levels are provided in Table S8.

Discussion

This study examined how inoculation messages, delivered in narrative and non-narrative formats, shape resistance to conflicting information and fatalistic beliefs. Relative to the control, both formats reduced fatalistic beliefs and heightened perceived efficacy, with modest effects on risk-reduction intentions. These results indicate the value of inoculation for countering the adverse effects of contradictory messages and underscore the

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flexibility of the theory, extending its relevance beyond the attitude protection of traditional persuasion to settings in which conflicting information erodes confidence in action.

A central implication of this study is that message format may matter less for inoculation than the preservation of its core structure. Although inoculation research has closely examined threat and refutational preemption, less attention has been given to whether these elements operate differently across narrative and non-narrative formats. This gap is notable given narrative's ambivalent role in resistance processes (Compton & Mason, 2020). Comparing two messages that shared the same inoculation logic but differed in format, we found that both elicited threat and supported resistance, and that neither consistently outperformed the other. Inoculation's effectiveness may thus depend less on format than on whether its essential elements are successfully conveyed.

The findings also refine how the role of narrative in inoculation should be understood. One concern is that narrative may undermine inoculation by reducing the critical processing that resistance requires (e.g., Compton & Ivanov, 2013; Compton & Mason, 2020). Yet the elevated post-attack counterarguing observed in the narrative condition suggests that whatever reduction in critical processing narrative may produce at initial inoculation exposure need not translate into weaker defensive responding when an attack later arrives. This pattern should nevertheless be interpreted with caution, as narrative inoculation did not produce significantly greater counterarguing than the non-inoculation control and thus cannot establish that narrative delivery enhanced resistance in absolute terms. What it does indicate is a format-specific difference between the two inoculation conditions: narrative delivery was more conducive than non-narrative delivery to this later resistance response. One possibility is that story-based presentation may help recipients encode or retain refutational material in ways that become useful at the point of attack (Appel & Richter, 2007). Although the present study cannot directly test this possibility, the results suggest that narrative is not inherently detrimental to inoculation and may, under some conditions, complement its defensive logic.

These results also speak to a related concern that narrative framing may narrow the umbrella of protection classic inoculation affords by binding resistance too tightly to particular characters or episodes, limiting its transfer to novel attacks (Compton & Mason, 2020). Although our design did not test breadth of protection across attack contexts, narrative inoculation was no less effective than its non-narrative counterpart on the outcomes assessed. Narrative delivery, in other words, appears not to have compromised inoculation's protective function, even if it conferred no consistent advantage. What may matter, then, is not whether a message is narrative, but how narrative elements are integrated with inoculation's core components in the service of defensive preparation.

The mediation results reinforce this interpretation. Transportation and identification did not account for the observed effects, consistent with evidence that these mechanisms are neither automatic nor uniformly beneficial in resistance contexts (Compton & Mason, 2020). They may be better understood as contingent audience responses than as defining criteria of narrative, with persuasive implications that vary across genre, length, medium, and character (Ratcliff & Sun, 2020); comparable work likewise finds that transportation and identification did not translate into stronger resistance to a delayed attack (Geegan et al., 2023). The absence of mediation here suggests that narrative engagement, at least as indexed by transportation and identification, may be insufficient on its own to strengthen inoculation. Its value may instead hinge on whether specific narrative features support inoculation's core architecture, for example through character development, emotional flow, or plot structure (Busselle & Bilandzic, 2009). Toward this theoretical direction, future research should therefore move beyond asking whether narrative is better or worse for inoculation to instead examine what kinds of narrative experiences meaningfully shape inoculation processes, identifying when narrative format alters not merely how inoculation looks, but how it works.

Beyond message formats, this study offers preliminary evidence that inoculation may serve both prophylactic and therapeutic functions depending on recipients' preexisting belief. In its classic prophylactic form, inoculation preserves resistance in advance of a challenge rather than initiating change (Compton, 2020; Wood, 2007). However, the present findings speak more to its therapeutic counterpart. Among participants higher in baseline fatalism, inoculation immediately reduced fatalistic beliefs relative to control, suggesting that the strategy may do more than shield existing beliefs. When conditions permit change, it may also disrupt maladaptive interpretations already in place and shift established beliefs (Ivanov et al., 2017; Wood, 2007). This extension of inoculation beyond protection to correction is particularly valuable given how difficult fatalistic beliefs can be to reverse once uncertainty has been internalized as evidence that prevention is futile.

Although the therapeutic evidence was clearer immediately after the intervention, the postattack interaction was nonsignificant. The findings therefore support immediate belief reduction but not a durable therapeutic effect. The inoculation effect also extended somewhat below the theoretical midpoint, reaching some participants whose baseline fatalism was comparatively low. This pattern is compatible with prophylactic potential, but an immediate reduction in fatalism does not, by itself, demonstrate protection against a subsequent attack. Whether inoculation also prophylactically protects these recipients from becoming more fatalistic following such a challenge remains a question for future research.

This tentative prophylactic–therapeutic functionality can be situated within prior work linking inoculation with efficacy. Pfau et al. (2009) held that threat highlights vulnerability, while refutational preemption builds confidence in defending attitudes. Jackson et al. (2015) extended this logic beyond attitudes, showing that inoculation can protect self-efficacy under challenge. Together, this work suggests that the present intervention may operate not only by prompting counterarguing but also by bolstering confidence in managing conflicting information—an important potential mechanism, since cancer fatalism is marked by diminished confidence in preventive action. This interpretation accords with emerging theory that inoculation is not purely cognitive: its threat component carries emotional implications and may interact with efficacy to shape resistance (Compton et al., 2022). More specifically, theorization linking inoculation and fear-and-efficacy processes suggests that resistance may be strongest when recipients perceive both vulnerability and coping capacity (Pfau et al., 2009). Where conflicting information evokes the uncertainty and helplessness that feed fatalistic

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interpretations, inoculation may help reframe anticipated challenges as manageable, potentially preserving a less fatalistic orientation while also enabling belief correction.

Taken together, the findings suggest that recipients' level of preexisting belief may condition their responses to inoculation, but they do not establish equally strong prophylactic and therapeutic effects across the full range of baseline beliefs. A single inoculation message may nevertheless reach audiences at different stages of preexisting belief without requiring entirely separate tailoring to each audience segment (Compton, 2020), although the uneven effects caution against assuming that one approach will benefit all recipients equally. More broadly, the findings illustrate the dual character of media effects as a source of maladaptive beliefs and a resource for resisting them, especially in communicating health, risk, and other domains marked by conflicting information. Given the modest effect sizes, single-exposure design, and nonsignificant postattack interaction, these observations remain preliminary. Future research should examine whether repeated exposure to inoculation messages produces more durable prophylactic and therapeutic effects.

Finally, although the inoculation affected fatalism and efficacy fairly consistently, it increased only intentions to consume fruits and vegetables and avoid alcohol in Phase 1 and to engage in physical activity in Phase 2. Inoculation, then, does not appear to enhance preventive intentions broadly; rather, it may influence the proximal cognitive antecedents of prevention more reliably than downstream intentions, which likely depend on the nature of the behavior, assessment timing, and context. Future work should examine when reduced fatalism and greater efficacy translate into durable intentions and, ultimately, behavioral change across domains, using behavioral follow-ups and repeated assessment over longer intervals.

The study has several limitations. It used a single message per condition and a shorter control message; this allowed an initial comparison of narrative and non-narrative inoculation but leaves open whether the findings are free of stimulus-specific confounds or generalizable beyond these messages. Future research should employ multiple messages (e.g., Niederdeppe et al., 2015) and length-matched controls while holding core inoculation elements constant to support stronger inferences about narrative form. In addition, the small effect sizes warrant caution regarding practical significance. Such modest effects may be expected from a brief, single-exposure intervention targeting change-resistant beliefs, but whether repeated exposure would produce more durable change in real-world settings remains a question the current design cannot resolve and future research should explore. Finally, this study did not comprehensively test mechanisms. Meaningful narratives may prompt self-reflection, reduce selective avoidance, satisfy autonomy needs, and lower reactance (Moyer-Gusé & Wilson, 2024); whether these processes explain narrative inoculation effects awaits investigation.

In conclusion, this study advances inoculation theory by extending its scope beyond the preservation of discrete attitudes to the modification of broader belief orientations and resistance to adverse media effects. In the context of conflicting cancer information, both narrative and non-narrative inoculation reduced fatalism and strengthened perceived efficacy, suggesting that inoculation's protective capacity may rest less on message format than on the effective delivery of threat and refutational preemption. At the same time, the narrative format's association with greater postattack counterarguing indicates that storytelling can complement, rather than compromise, defensive processing. Additionally, the findings point to possible prophylactic and therapeutic applications of inoculation, highlighting its potential as a flexible strategy for fostering resilience amid informational uncertainty.

Supplementary material

Supplementary material is available at *Journal of Communication* online.

Data availability

The data underlying this article will be shared upon reasonable request to the corresponding author.

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Conflicts of interest

None declared.

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